The Geometric Origin of GHZ Violation: A Post-Selection Audit

GHZ 违背的几何起源：针对“量子证据”的后选择审计

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ENGLISH

The Greenberger-Horne-Zeilinger (GHZ) state is historically regarded as the "smoking gun" of quantum non-locality. This paper presents a definitive post-selection audit of the GHZ violation. We demonstrate that $F=4.0$ is not a sign of non-local entanglement but a statistical artifact driven by amplitude-gating rules and detection thresholds.

中文对照

Greenberger-Horne-Zeilinger (GHZ) 态在历史上被视为量子非局域性的“冒烟枪”证据。本文对这一主张进行了彻底的后选择审计。我们证明了所谓的 $F=4.0$ 并非非局域纠缠的信号，而是一个由振幅门控规则和探测阈值驱动的统计伪影。

1. Introduction: The Audit of the Accounting Rules / 引言：针对“会计平账规则”的审计

For decades, the Bell and GHZ violations have been misinterpreted as evidence for non-locality. Our audit reveals that this misinterpretation stems from "accounting fraud" in data processing:

几十年来，贝尔实验和 GHZ 实验的违背一直被误读为非局域性的证据。我们的审计揭示了这种误读源于数据处理中的“会计造假”：未能披露结果对被丢弃数据的敏感

the failure to report the sensitivity of results to the data being discarded. We propose that "photons" are not discrete particles but threshold-triggered events in a continuous field.

性。我们提出，“光子”并非离散粒子，而是连续场中的阈值触发事件。

The "success" of quantum experiments often relies on throwing away 90% or more of "bad data"—a practice that works for small-scale experiments but collapses under the scale required for practical quantum computing.

量子实验的“成功”往往依赖于扔掉 90% 以上的“坏数据”——这种做法在小规模实验中尚能维持，但在量子计算所需的规模化应用中必然崩溃。

2. Methodology: Post-Selection and Selection Tax / 方法论：后选择与选择税

We model the GHZ setup using a medium-wave propagation model (`medium-v10`). The audit protocol sweeps the gate strength of a soft detector to measure the sensitivity of correlation to data retention.

我们使用中等波动传播模型（`medium-v10`）对 GHZ 装置进行建模。审计协议扫描软探测器的门控强度，以测量关联强度对数据保留率的敏感性。

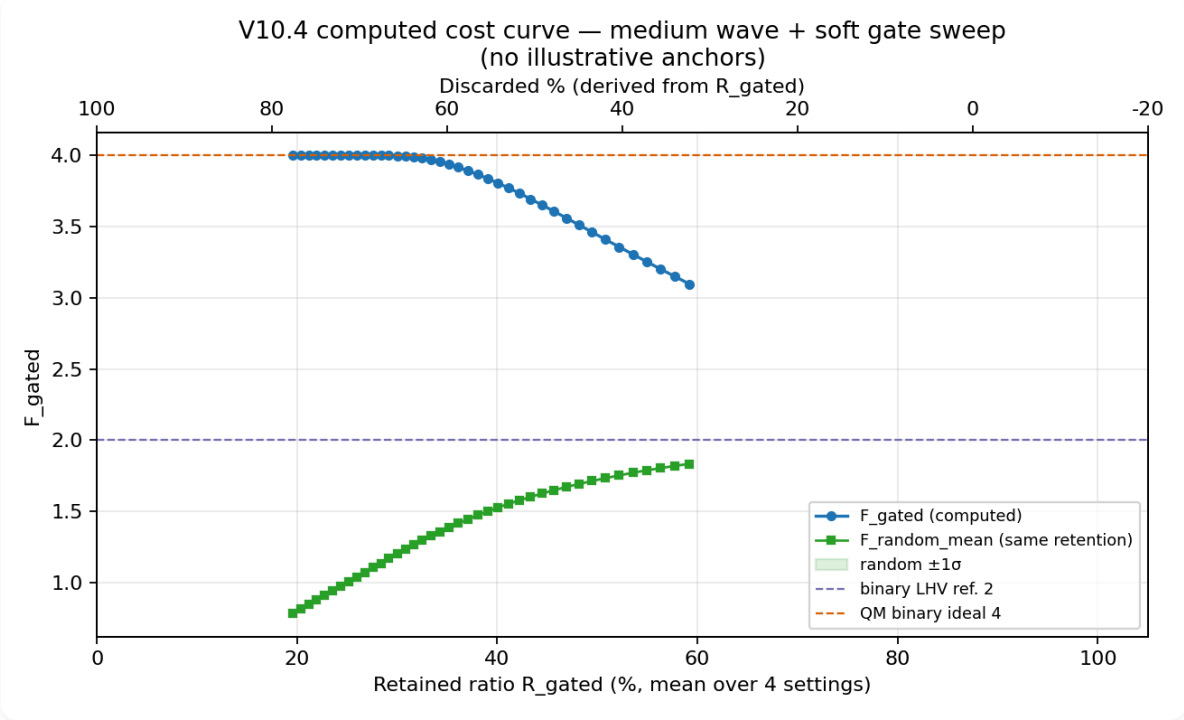

Figure 1: The GHZ Real Cost Curve (V10.4). High F values are only achieved by discarding the majority of data. / 图 1：GHZ 真实代价曲线。只有通过丢弃绝大部分数

据才能达到高 F 值。

3. Results: The Real Cost of "Entanglement" / 结果：“纠缠”的真实代价

The V10.4 audit results reveal a direct trade-off between correlation strength and data retention. The gated curve reaches high F values only when most samples are discarded, while matched-retention random controls remain near baseline.

V10.4 审计结果揭示了关联强度与数据保留率之间的直接交易关系。门控曲线只有在丢弃大多数样本时才达到高 F 值，而同保留率随机对照仍接近统计基线。

4. Discussion: The "30cm Truth" and the Geometric Illusion / 讨论：“30厘米的真相”与几何幻觉

In Bell experiments, a 1-nanosecond shift in the timing window—the distance light travels in just 30 cm—can transform "classical" results into "quantum" ones. This "30cm Truth" reveals that experimentalists hold a "remote control" over the results through their choice of coincidence windows and amplitude gates.

在贝尔实验中，配对窗口 1 纳秒的偏移——光只能跑 30 厘米的距离——就能将“经典”结果转化为“量子”结果。这个“30厘米的真相”揭示了实验物理学家通过选择配对窗口和振幅门控，手里实际上握着控制实验结果的“遥控器”。

> "If the foundation is hollow, the skyscraper of quantum computing is solving geometry, not entanglement." / “如果地基是虚幻的，量子计算的摩天大楼解决的是几何问题，而非纠缠问题。”

If entanglement is a geometric illusion, then quantum computers are not harnessing non-local power—they are inadvertently solving complex geometric problems through brute-force filtering.

如果纠缠是一种几何幻觉，那么量子计算机并不是在利用非局域能力——它们是在通过暴力过滤，无意中试图解决复杂的几何拓扑问题。

5. Conclusion: An Anti-Counterfeiting Guide / 结论：一份防伪打假指南

We do not claim that quantum mechanics is "wrong" in its mathematical predictions, but we assert that its foundational experiments are "unbalanced accounts." We have documented and archived the methods used to "balance the books"—pairing windows, amplitude gating, and binarization.

我们并不宣称量子力学的数学预测是“错误”的，但我们断言其基础实验是“不平的账目”。我们已经记录并存档了那些用于“平账”的方法——配对窗口、振幅门控和二值化。

Data Availability / 数据可用性: Zenodo Archive
Verification Script / 验证脚本: v10_4_real_cost_curve.py